

팍리스 실무형 일경험 프로그램

SDV 아키텍처 기반 자율주행 통합 시스템 구현 실습



1일차

05 D3-G 환경 구성

Telechips TOPST

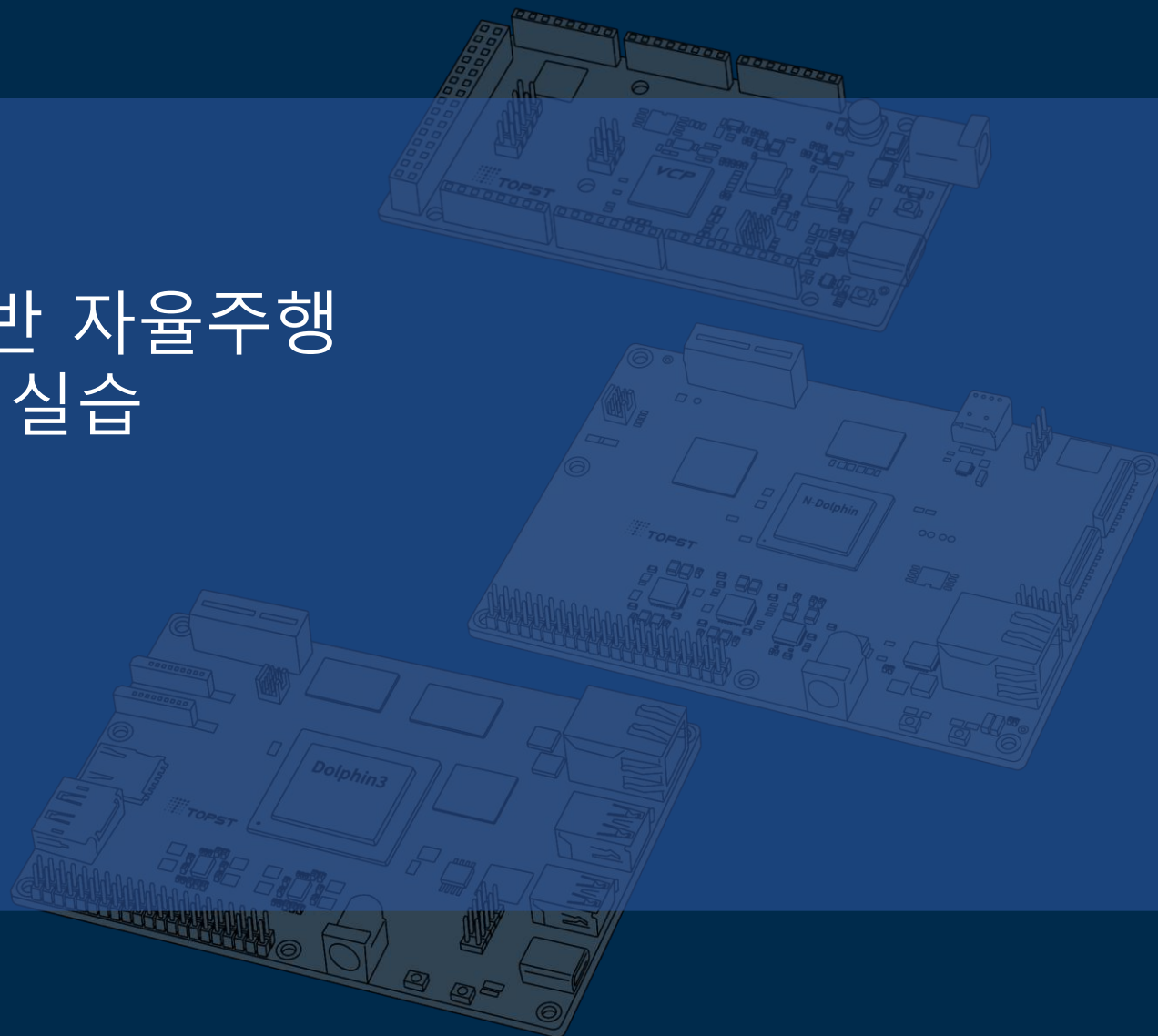


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D3-G Specification

Part Name	D3-G
CPU	Arm® Cortex-A72 1.6 GHz, Quad
CPU	Arm® Cortex-A53 1.45 GHz, Quad
CPU	Arm® Cortex-R5
GPU	168 GFLOPS, IMG PowerVR 9XTP
Memory	4 GB (2 GB 32-bit LPDDR4 x 2)
Storage	microSD Card (Not included), 8 GB eMMC 5.1 Flash Storage
Automotive Network	CAN 3-channel
Network	Gigabit Ethernet
Camera	MIPI CSI-2 2-lane, MIPI CSI-2 4-lane
Display	DisplayPort 1.4
USB	USB 2.0 Host, USB 2.0 DRD, USB 3.0 DRD
Others	PCIe Gen3, UART/JTAG debugger port, I2S, I2C, SPI, PWM
Power	5V/3A
OS	Linux
Mechanical	120 mm x 90 mm

- D3-G 는 3개의 코어 존재
 - Cortex-A72
 - Cortex-A53
 - Cortex-R5
- 본 실습에선 **A72 코어와 R5 코어만**을 사용
 - A72 – Ubuntu Weston Desktop 설치
 - R5 – FreeRTOS 설치

D3-G Yocto Build

```
$ mkdir -p Work/d3-g-yocto  
$ cd Work/d3-g-yocto  
$ repo init -u https://github.com/topst-development/manifests.git -b release/1.1.0 -m  
linux\_yp4.0\_topst.xml
```

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ repo init -u https://github.com/topst-development/manifests.git -b release/1.1.0  
-m linux_yp4.0_topst.xml  
Downloading Repo source from https://gerrit.googlesource.com/git-repo  
remote: Finding sources: 100% (4/4)  
remote: Total 4 (delta 3), reused 4 (delta 3)  
  
... A new version of repo (2.54) is available.  
... New version is available at: /home/kypark/Work/d3-g-yocto/.repo/repo/repo  
... The launcher is run from: /usr/bin/repo  
!!! The launcher is not writable. Please talk to your sysadmin or distro  
!!! to get an update installed.  
  
Your identity is: kiyong.park <kiyong.park@telechips.com>  
If you want to change this, please re-run 'repo init' with --config-name  
  
repo has been initialized in /home/kypark/Work/d3-g-yocto  
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ |
```

D3-G Yocto Build

```
$ repo sync
```

```
$ ls
```

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ repo sync
```

```
... A new version of repo (2.54) is available.  
... New version is available at: /home/kypark/Work/d3-g-yocto/.repo/repo/repo  
... The launcher is run from: /usr/bin/repo  
!!! The launcher is not writable. Please talk to your sysadmin or distro  
!!! to get an update installed.
```

```
Syncing: 25% (3/12), done in 32.045s
```

```
Syncing: 100% (12/12) 0:48 | 1 job | 0:14 meta-openembedded @ poky/meta-openembeddedrepo sync has finished successfully.
```

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ ls
```

```
easy-setup.sh mktcimg poky stitch-fai-ai.sh stitch-fai-d3.sh tools
```

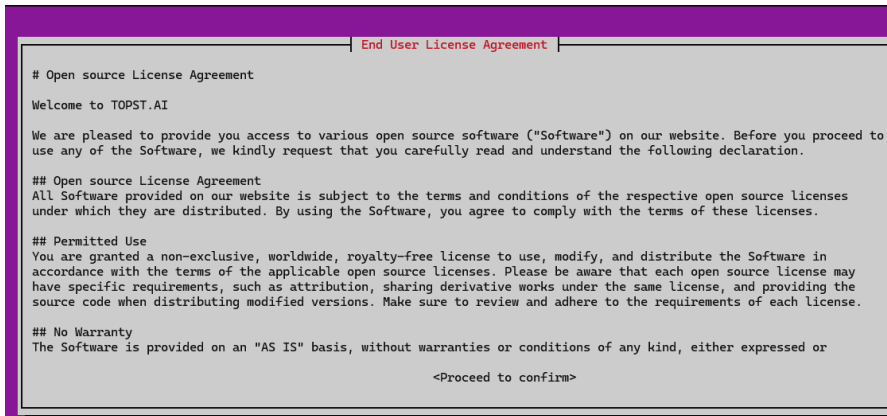
```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ |
```

D3-G Yocto Build

topst-build.sh 실행

```
$ ./easy-setup.sh
```

스크롤 내리기 -> 방향키(->) 클릭 -> 엔터 -> accept 엔터



```
End User License Agreement

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## No Warranty
The Software is provided on an "AS IS" basis, without warranties or conditions of any kind, either expressed or

<Proceed to confirm>
```



D3-G Yocto Build

topst-build.sh 실행

\$ source poky/meta-topst/topst-build.sh -> 2번 머신 선택 (2 입력)

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto$ source poky/meta-topst/topst-build.sh
Choose MACHINE
 1. ai-g-topst
 2. d3-g-topst-main
 3. d3-g-topst-sub
 4. d5-g-topst-main
 5. d5-g-topst-sub
select number(1-5) => 2
machine(d3-g-topst-main) selected.
You had no conf/local.conf file. This configuration file has therefore been
created for you from /home/kypark/Work/d3-g-yocto/poky/meta-topst/template/d3-g-topst-main/local.conf.sample
You may wish to edit it to, for example, select a different MACHINE (target
hardware). See conf/local.conf for more information as common configuration
options are commented.

You had no conf/bblayers.conf file. This configuration file has therefore been
created for you from /home/kypark/Work/d3-g-yocto/poky/meta-topst/template/d3-g-topst-main/bblayers.conf.sample
To add additional metadata layers into your configuration please add entries
to conf/bblayers.conf.

The Yocto Project has extensive documentation about OE including a reference
manual which can be found at:
  https://docs.yoctoproject.org

For more information about OpenEmbedded see the website:
  https://www.openembedded.org/

Yocto Project common targets are:
  core-image-minimal
  core-image-sato
```

D3-G Yocto Build

Yocto 빌드 수행

```
$ bitbake telechips-topst-image
```

```
kypark@TC-E-20-0417:~/Work/d3-g-yocto/build/d3-g-topst-main$ bitbake telechips-topst-image
NOTE: Your conf/bblayers.conf has been automatically updated.
WARNING: You are running bitbake under WSLv2, this works properly but you should optimize your VHDX file eventually to a
void running out of storage space
Loading cache: 100% | ETA: --:--:--
Loaded 0 entries from dependency cache.
Parsing recipes: 100% |#####| Time: 0:00:24
Parsing of 2382 .bb files complete (0 cached, 2382 parsed). 3770 targets, 325 skipped, 0 masked, 0 errors.
WARNING: No recipes in default available for:
/home/kypark/Work/d3-g-yocto/poky/meta-topst/recipes-connectivity/nanomsg/nng_1.%.bbappend
NOTE: Resolving any missing task queue dependencies

Build Configuration:
BB_VERSION           = "2.0.0"
BUILD_SYS            = "x86_64-linux"
NATIVESBSTRING       = "ubuntu-22.04"
TARGET_SYS           = "aarch64-telechips-linux"
MACHINE              = "d3-g-topst-main"
DISTRO               = "poky-telechips-systemd"
DISTRO_VERSION       = "4.0.17"
TUNE_FEATURES        = "aarch64 armv8a crc cortexa72"
TARGET_FPU           = ""
LINUX_VERSION        = "5.10.205"
KBUILD_DEFCONFIG      = "d3_g_topst_defconfig"
TCC_BSP_FEATURES     = "audio display graphics camera drm network multimedia_vpu_v3"
IMAGE_FEATURES       = " u-boot-net debug-tweaks ssh-server-openssh package-management"
GCCVERSION           = "arm-11.2"
GLIBCVERSION         = "2.35"
TOPST_FEATURES       = " multimedia support-4k-video network support-tty-console support-bt-usb support-nf-docker suppor
t-swap"
```


D3-G Yocto Build

Yocto 빌드 결과 dir 설명

- Kernel : d3-g-topst-main-tc-boot-5.10.205-r0.img, Image--5.10.205-r0-d3-g-topst-main-20250929060303.bin
- Kernel Modules : modules--5.10.205-r0-d3-g-topst-main-20250929060303.tgz
- Device tree bin : tcc8050-topst-d3-g--5.10.205-r0-d3-g-topst-main-20250929060303.dtb
- Rootfs : telechips-topst-image-d3-g-topst-main-20250929060303.rootfs.ext4

```
kypark@TOPST-Build-SVR:~/Work/d3g-github-recent/build/d3-g-topst-main/tmp/deploy/images/d3-g-topst-main$ pwd
/home/kypark/Work/d3g-github-recent/build/d3-g-topst-main/tmp/deploy/images/d3-g-topst-main
kypark@TOPST-Build-SVR:~/Work/d3g-github-recent/build/d3-g-topst-main/tmp/deploy/images/d3-g-topst-main$ ls
ca72_bl3-d3-g-topst-main-1.0-r0.rom          modules-d3-g-topst-main.tgz
ca72_bl3-d3-g-topst-main.rom               tc-boot-d3-g-topst-main.img
ca72_bl3.rom                               tc-boot.img
d3-g-topst-main-tc-boot-5.10.205-r0.img     tcc8050-topst-d3-g--5.10.205-r0-d3-g-topst-main-20250929060303.dtb
Image                                       tcc8050-topst-d3-g-d3-g-topst-main.dtb
Image--5.10.205-r0-d3-g-topst-main-20250929060303.bin
Image-d3-g-topst-main.bin                  tcc8050-topst-d3-g.dtb
Image-initramfs--5.10.205-r0-d3-g-topst-main-20250929060303.bin
Image-initramfs-d3-g-topst-main.bin         telechips-topst-image-d3-g-topst-main-20250929060303.rootfs.ext4
initramfs-topst-image-d3-g-topst-main-20250929060303.rootfs.cpio
initramfs-topst-image-d3-g-topst-main-20250929060303.rootfs.manifest
initramfs-topst-image-d3-g-topst-main-20250929060303.testdata.json
initramfs-topst-image-d3-g-topst-main.cpio  telechips-topst-image-d3-g-topst-main-20250929060303.rootfs.manifest
initramfs-topst-image-d3-g-topst-main.manifest
initramfs-topst-image-d3-g-topst-main.testdata.json
modules--5.10.205-r0-d3-g-topst-main-20250929060303.tgz
kypark@TOPST-Build-SVR:~/Work/d3g-github-recent/build/d3-g-topst-main/tmp/deploy/images/d3-g-topst-main$
```

D3-G Yocto 이미지

fwdn 이미지 추출

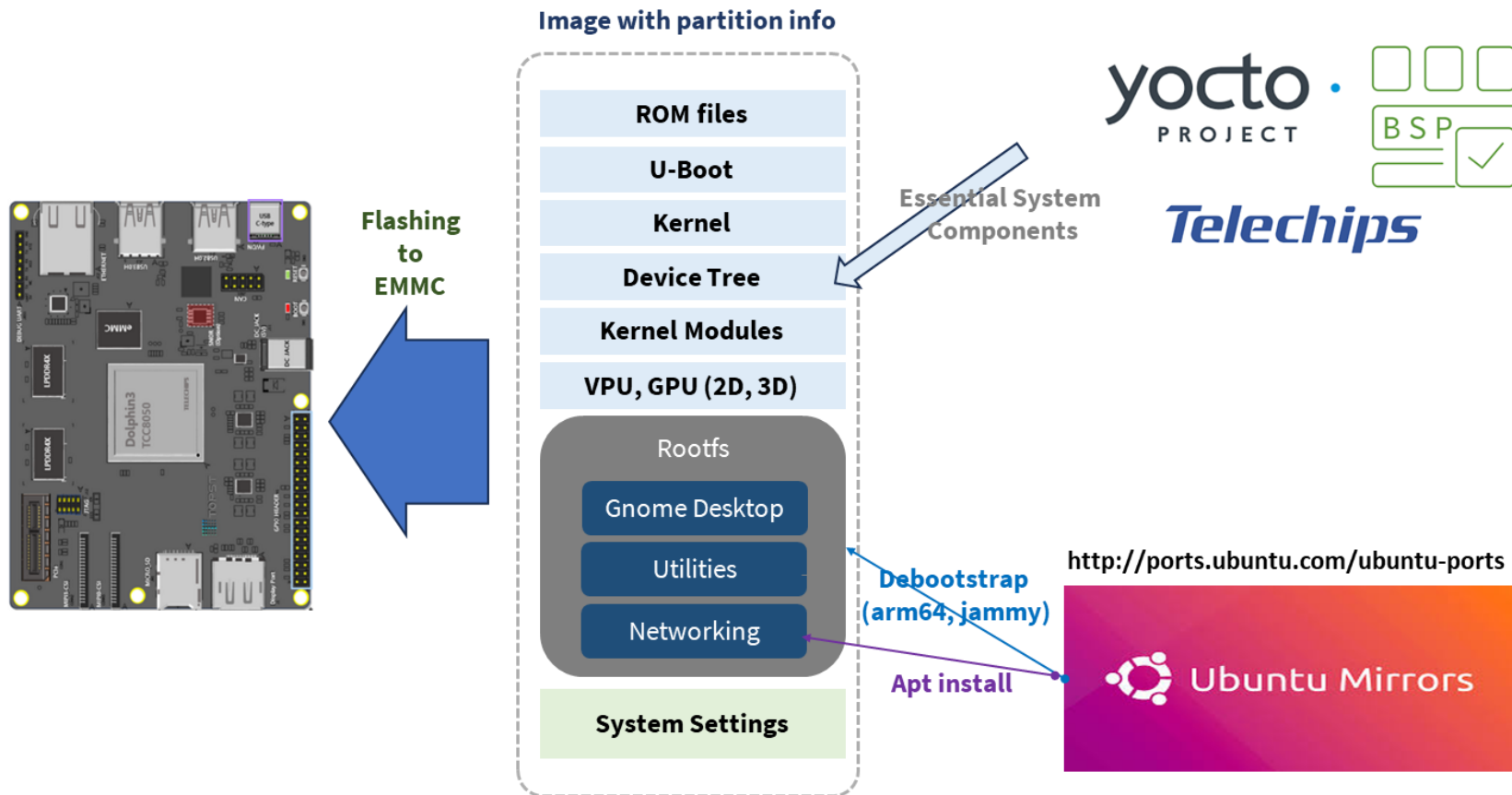
```
$ cd ~/Work/d3-g-yocto
```

```
$ ./stitch-fai-d3.sh -f => output_d3g.fwdn.zip 파일 생성 (fw 설치 위한 필요 파일 압축본)
```

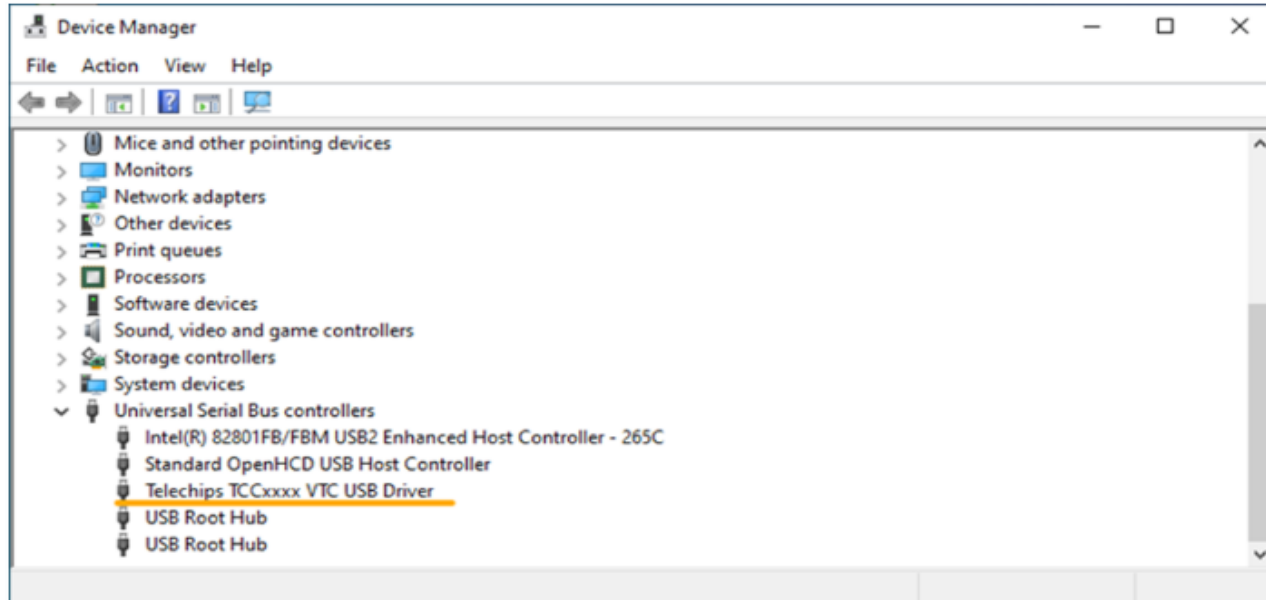
```
kypark@T0PST-Build-SVR:~/Work/d3g-github-recent$ ls -al
total 216380
drwxrwxr-x  7 kypark kypark      4096 Nov  3 09:28 .
drwxrwxr-x 60 kypark kypark      4096 Oct 16 13:51 ..
drwxrwxr-x  3 kypark kypark      4096 Sep 29 15:02 build
lrwxrwxrwx  1 kypark kypark        19 Sep 29 15:01 easy-setup.sh -> tools/easy-setup.sh
-rw-rw-r--  1 kypark kypark         0 Sep 29 15:02 .eula_accepted
drwxrwxr-x  4 kypark kypark      4096 Sep 29 15:01 mktcimg
-rw-rw-r--  1 kypark kypark 221536745 Nov  3 09:28 output_d3g.fwdn.zip
drwxrwxr-x 20 kypark kypark      4096 Sep 29 15:01 poky
drwxrwxr-x  7 kypark kypark      4096 Sep 29 15:01 .repo
lrwxrwxrwx  1 kypark kypark        22 Sep 29 15:01 stitch-fai-ai.sh -> tools/stitch-fai-ai.sh
lrwxrwxrwx  1 kypark kypark        22 Sep 29 15:01 stitch-fai-d3.sh -> tools/stitch-fai-d3.sh
drwxrwxr-x  4 kypark kypark      4096 Sep 29 15:01 tools
kypark@T0PST-Build-SVR:~/Work/d3g-github-recent$
```

D3-G Ubuntu 이미지

Ubuntu Image 다운로드 : [D3-G Gnome image](#)



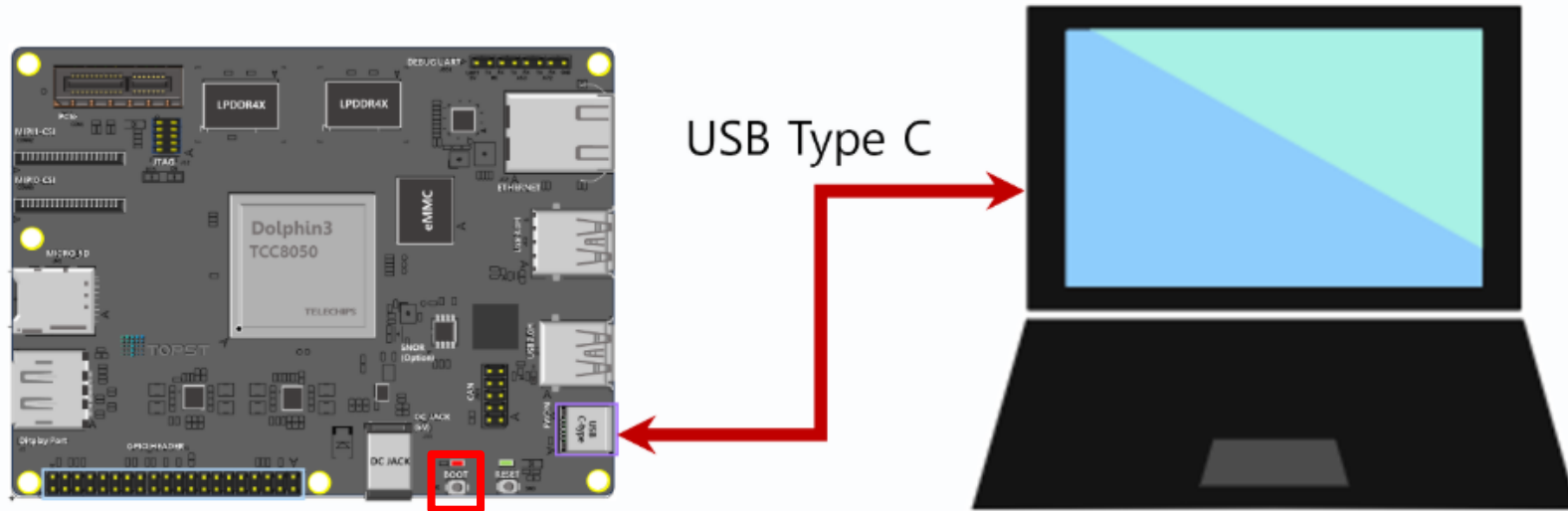
D3-G 환경 설정(공통)



• Install VTC Driver

- 펌웨어 다운로드를 위해 VTC(Vendor Telechips Certification) 드라이버 설치 필수
- 관리자 권한으로 실행: [VTC 드라이버](#)
- 드라이버 설치 후 FWDN 모드에서 USB 연결 후 설정 확인

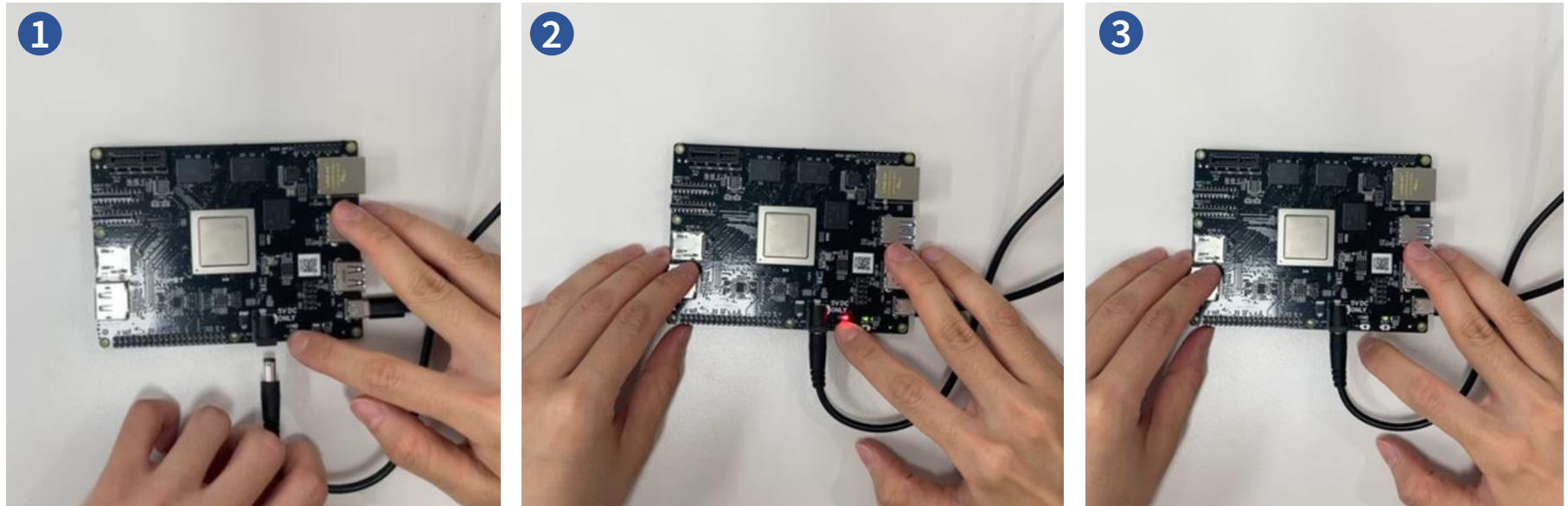
D3-G 환경 설정 (공통)



- **Connect D3-G Board and Host PC with USB Boot Mode**

- USB Boot Mode로 들어가기 위해 FWDN 스위치를 누른 상태에서 전원 케이블을 D3-G 보드에 연결
- USB Type-C 케이블을 D3-G 보드 및 Host PC의 USB Type-C FWDN 포트에 연결
- D3-G : **C 케이블** & Host PC : **USB 케이블**

D3-G 환경 설정 (공통)



- **Boot Mode 진입**

- BOOT 스위치를 누른 채로, 전원 플러그 연결(BOOT Switch LED 상태 : ON)
- 이후 스위치에서 손가락을 떼면 진입 완료(BOOT Switch LED 상태 : OFF)

D3-G A72 이미지 업로드 (공통)

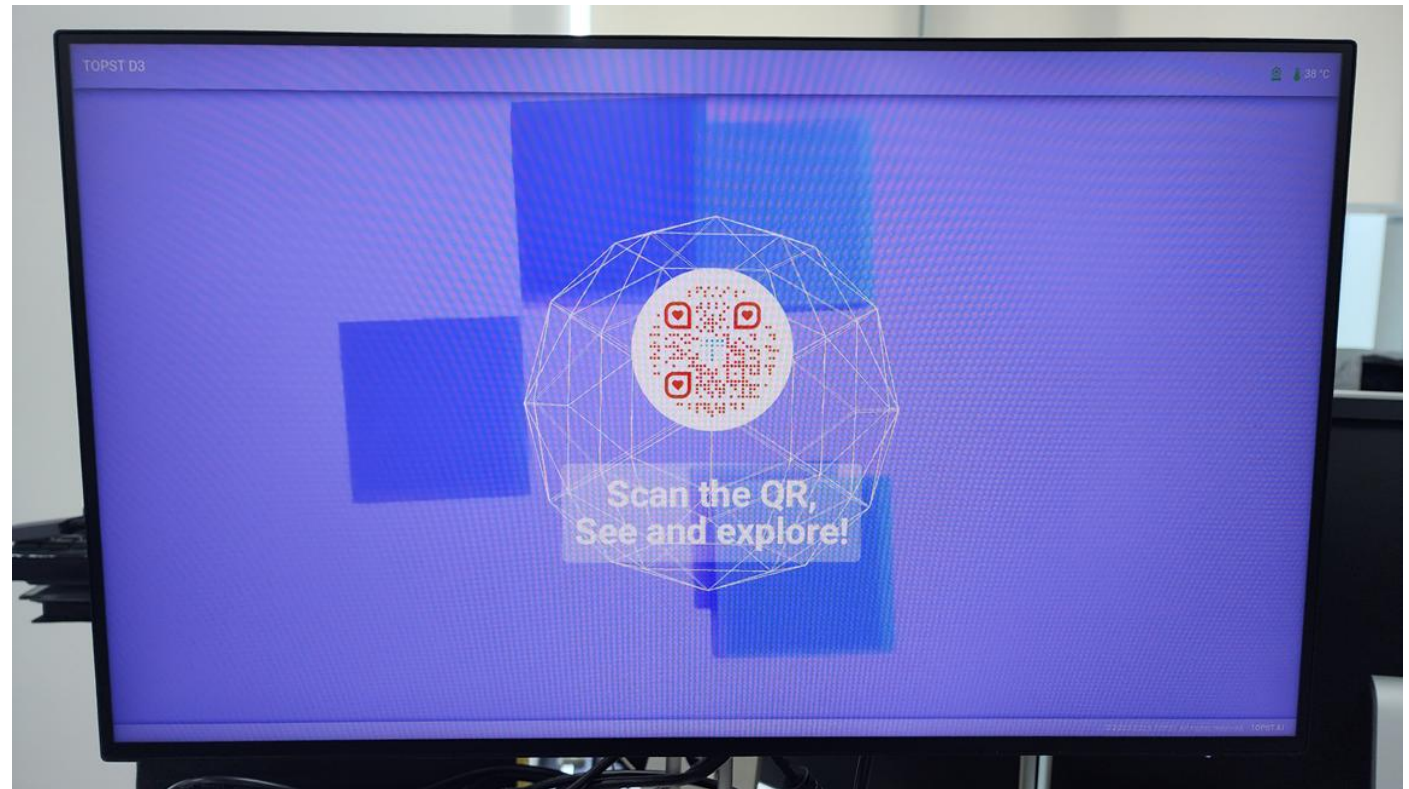


• FWDN (D3-G Ubuntu Weston/Gnome Desktop)

- 제공된 d3g-ubuntu zip 파일을 PC Windows 혹은 PC ubuntu에 다운로드 후 압축 해제
- Ex) Windows
 - 받은 폴더 내에서 마우스 우클릭> 터미널에서 열기(명령프롬프트) 실행
 - fwdn.bat (Windows 배치 파일)실행
 - fwdn.bat은 펌웨어를 자동으로 다운로드 하는 실행 파일
 - \$.\fwdn.bat

D3-G A72 이미지 (Yocto Weston)

- 설치가 완료되면 전원 재인가 or reset 버튼 누르기



Demo Web-site on localhost

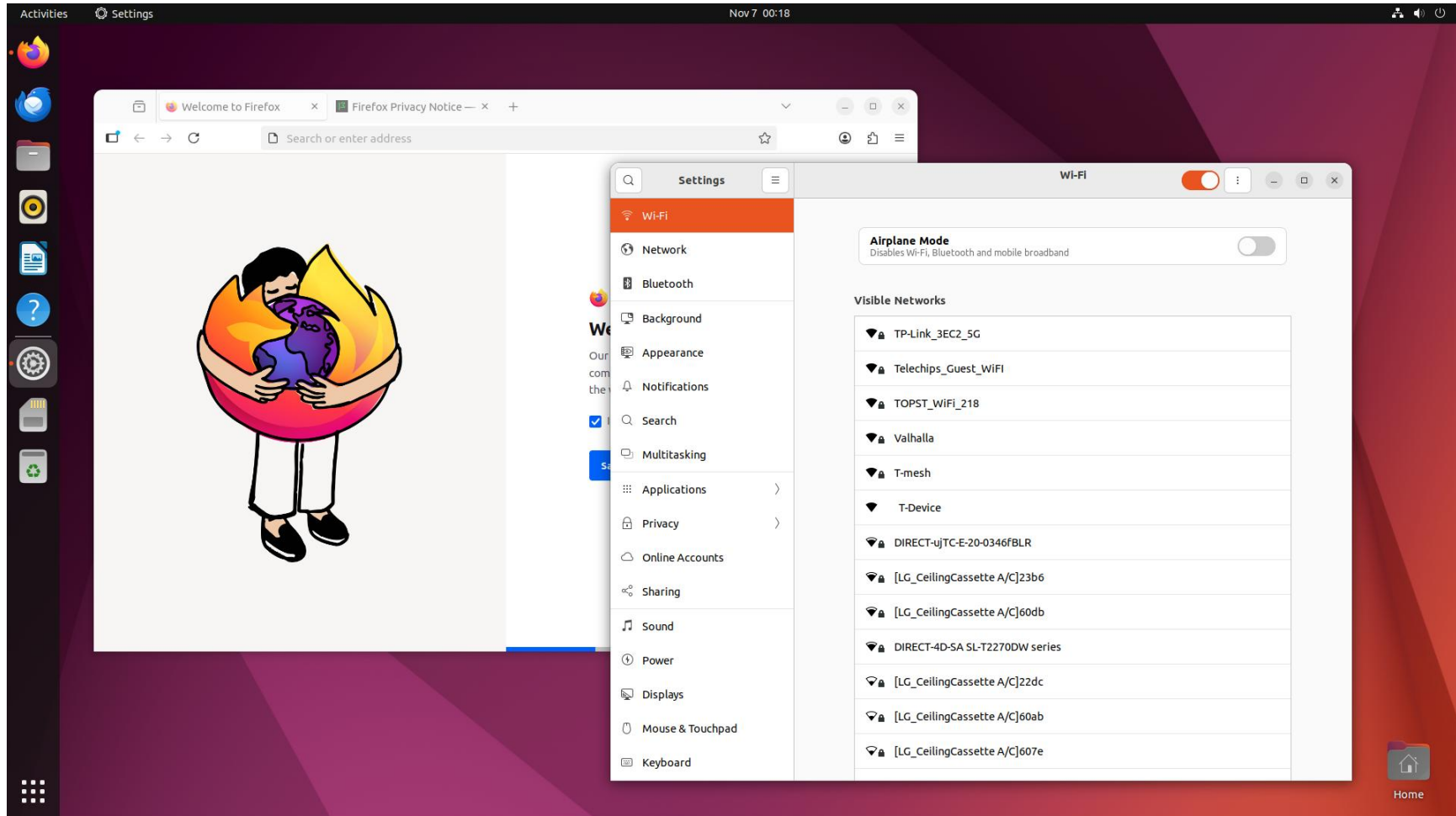
COG browser

WPE Webkit

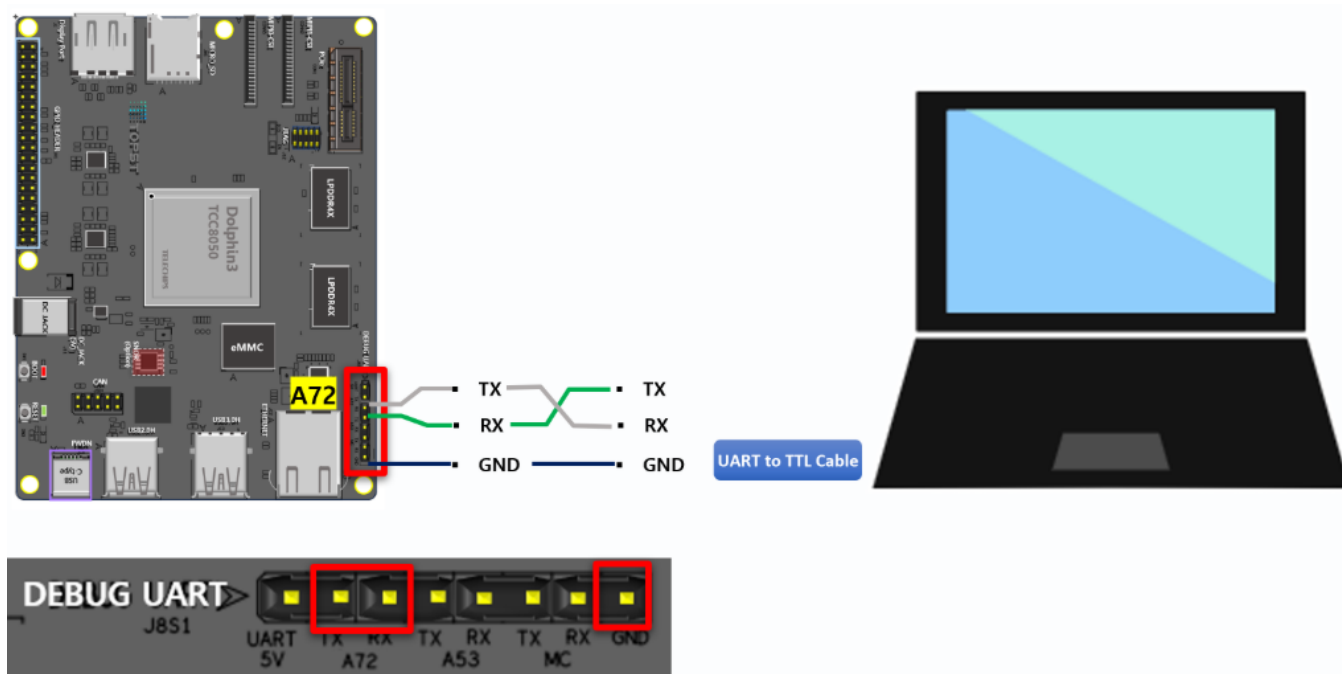
WPE Backend

D3-G system

D3-G A72 이미지 (Ubuntu Gnome)

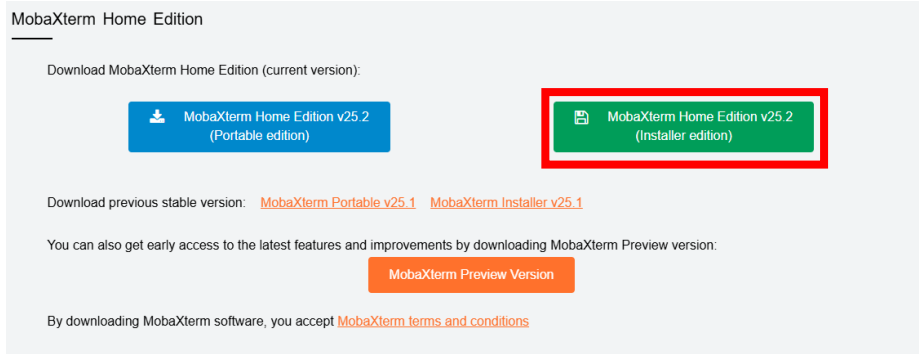


D3-G A72 ↔ PC 통신



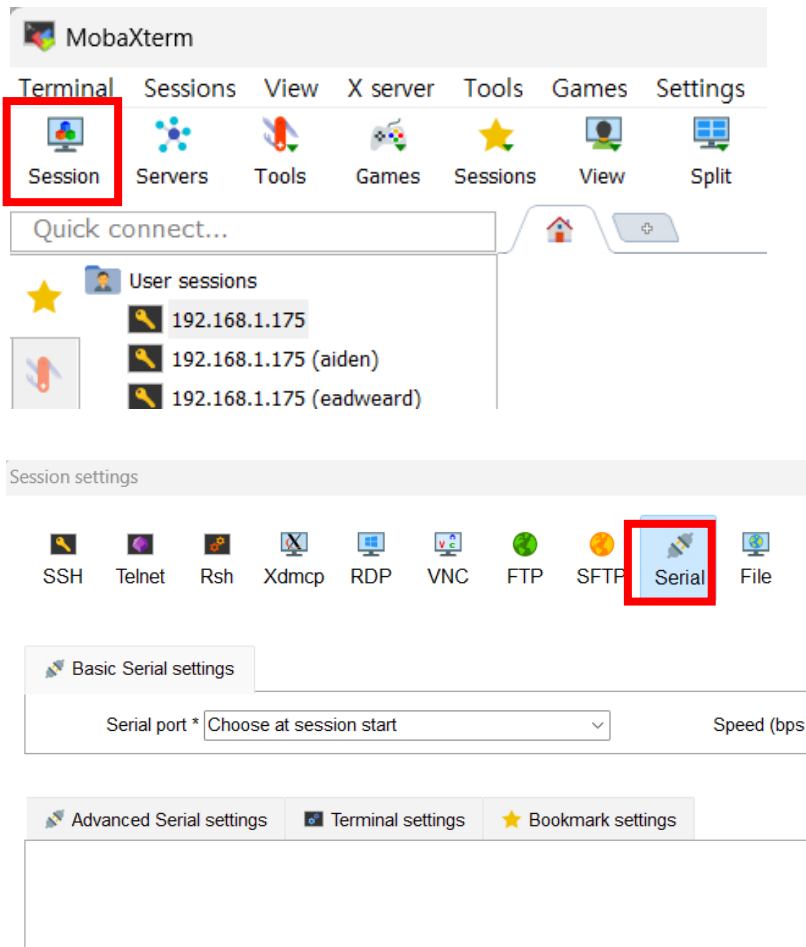
- **Connect D3-G A72 and Host PC with UART**
 - D3-G A72 코어와 PC 간 USB-to-TTL cable을 이용하여 연결
 - 검정 케이블은 UART 핀의 **GND**에 연결
 - 흰 케이블은 UART 핀의 **TX** 핀에 연결 & 녹색 케이블은 UART 핀의 **RX** 핀에 연결

D3-G A72 ↔ PC 통신

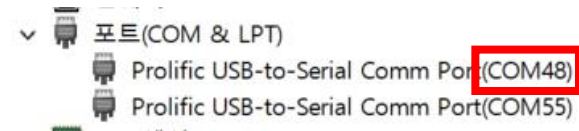


- **Connect D3-G Board and Host PC with UART**
 - Serial port driver 설치
 - [CP210x Universal Windows Driver](#)
 - [PL2303_prolific Driver](#)
 - **Terminal emulator 설치 (MobaXterm)**
 - [MobaXterm Xserver with SSH, telnet, RDP, VNC and X11 - Download](#) - 링크 접속
 - **압축 해제**
 - 설치한 MobaXterm_Installer_v25.2 압축 해제
 - MobaXterm_installer_25.2 실행파일 실행
-  MobaXterm_installer.dat
-  MobaXterm_installer_25.2
- Next 두는 우 실시 진행

D3-G A72 ↔ PC 통신



- 윈도우 검색창에 MobaXterm 검색 후 실행
- 오른쪽 상단에 Session 버튼 클릭
- Serial 클릭
- 장치 관리자를 통해 port 확인



D3-G A72 ↔ PC 통신

Basic Serial settings

Serial port * COM4 (Prolific USB-to-Serial Comm Port(CO... Speed (bps) * 115200

Advanced Serial settings Terminal settings Bookmark settings

Serial engine: PuTTY (allows manual COM port setting)

Data bits 8

Stop bits 1

Parity None

Flow control None

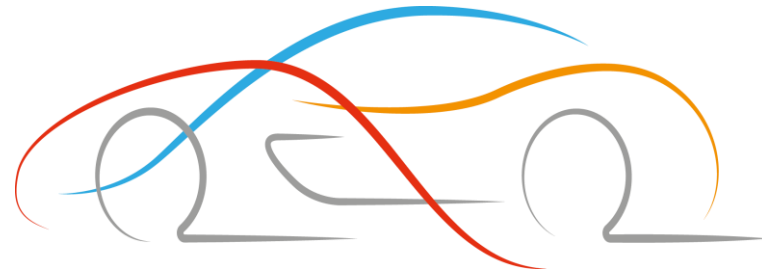
Reset defaults

Execute macro at session start: <none>

If you need to transfer files (e.g. router configuration file), you can use MobaXterm embedded TFTP server

"Servers" window --> TFTP server

- Serial port에서 연결된 포트 설정
 - Speed 115200 설정
- Advanced Serial settings 탭 선택
 - Flow control에서 None 선택
- ok눌러 보드 접속
- ID : root / PW : root 입력



Thank You